

Quiz - Thermodynamics

Important Instructions

- a) A question may have multiple correct answers. b) An answer is deemed correct if all the correct answers are marked. c) Marks will be deducted for each wrong answer.

Time: 50 Minutes

Total Marks: 40

1. Identify the incorrect statement. **Ans: b**
 - a) $Tds = du + pdv$ is valid for both reversible and irreversible processes
 - b) $Tds = du + pdv$ is valid for any simple system, but the working substance must be a pure substance.
 - c) $Tds = du + \delta w$ is valid for closed stationary systems undergoing at least an internally reversible process.
 - d) $\delta q = du + pdv$ is valid for a simple compressible closed system or a steady flow open system undergoing reversible processes.
2. Identify the correct statement. **Ans: c**
 - a) In the p-h plot of a pure substance in the superheated zone, the isotherms become parallel to isenthalpic lines as the degree of superheat decreases.
 - b) On the h-s plot of a pure substance, over the vapor dome enthalpy is maximum at the critical point.
 - c) In the two-phase zone of a h-s plot, the isobars are straight lines.
 - d) On a T-s plot, the isochores have a lower slope than isobars.
3. Identify the incorrect statement. **Ans: d**
 - a) If an ideal gas with constant specific heats undergoes an irreversible process from state 1 to state 2 such that $s_1 = s_2$ then $p_1 v_1^n = p_2 v_2^n$ where "n" is the ratio of the specific heats.
 - b) A irreversible cycle is possible with net work in and net heat out while the system interacts with a single reservoir.
 - c) All internally reversible heat engines may have different efficiencies while operating between the same temperature limits.
 - d) A heat engine is possible with the system taking heat from a low-temperature source and rejecting heat to a high-temperature sink.
4. If 1 and 2 are the states before and after an adiabatic throttling of a constant density liquid, then which of the following is correct? "u" is the specific internal energy. **Ans: b**
 - a) $u_2 = u_1$
 - b) $u_2 > u_1$
 - c) $u_2 < u_1$
 - d) Can not be predicted
5. Identify the correct statement. **Ans: d**
 - a) Below the triple point temperature, the liquid phase cannot exist.
 - b) The triple phase state either be a point or a line on any thermodynamic plot.
 - c) The degrees of freedom at triple point are three.
 - d) At triple point the enthalpy of sublimation is equal to the sum of enthalpy of fusion and vaporization

6. Identify which of the following statements is not true for an ideal gas. **Ans: c**
- For ideal gas undergoing constant pressure process $du = c_v dT$
 - For ideal gas undergoing reversible isothermal process in a simple stationary closed system $Tds = \delta w$
 - For an ideal gas in a simple steady flow system undergoing a reversible isothermal process $Tds = \delta w$
 - For an ideal gas, entropy is a function of both temperature and pressure
 - For an ideal gas, the compressibility factor is 1 for all temperatures and pressure
7. Which of the following is a pure substance? **Ans: b**
- A mixture of liquid and gaseous air.
 - Gaseous combustion product
 - Moist air (dry air + water vapour)
 - A mixture of water and ammonia
8. Identify the incorrect statement. **Ans: d**
- A refrigerator undergoes net negative work and net negative heat interactions.
 - The vapour compression refrigeration cycle is neither externally reversible nor internally reversible.
 - The COP of a refrigerator can be less than 1, but the COP of a heat pump must be more than one.
 - In an irreversible refrigerator, the entropy change of refrigerant during every cycle is positive.
9. A fully reversible heat engine receives **220,000 kJ/h** from a source at **350°C** and rejects **130,000 kJ/h** to another cold reservoir. Which of the following statements is not correct?
Ans: c
- The cold reservoir temperature is 95°C and the irreversibility associated with the engine is 0.
 - The efficiency is 40.9%, and the net power output is 90,000 kJ/h
 - According to the 2nd law, if the cold reservoir temperature was 0°C, the heat rejection rate of the reversible heat engine might have been 0 kJ/h.
 - If the same heat engine is reversed as a heat pump, the COP will be 2.44
10. Carbon dioxide is compressed to a smaller volume in a polytropic process with **n = 1.2**. in a steady flow compressor. Which of the following statements is not correct? The isentropic index for Carbon dioxide is **1.28**. **Ans: a**
- $v_{exit,polytropic} > v_{exit,isentropic}$
 - $|Q_{isothermal}| > |Q_{polytropic}|$
 - $|w_{polytropic}| < |w_{isentropic}|$
 - $|w_{polytropic}| > |w_{isothermal}|$
11. Which of the following statements regarding the reversible cycles A and B is correct?
Ans: a

a) The efficiency of B is greater than that of A b) Heat input to cycle A is the same as that of B c) A has two adiabatic processes. The same is the case with B d) The mean temperature of heat rejection for cycle A is lower than that of B	
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12. Figure shows isobaric, isochoric, and isothermal processes on the T-s plot. Identify the correct statement. **Ans: c**

a) "a" is an isobar b) "b" is an isochore c) "c" is an isobar d) "d" is an isobar e) can only be predicted if the vapor is assumed to be an ideal gas	
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13. A room is heated with a **2000 W** electric heater. How much power can be saved if a heat pump with a COP of **2.5** is used instead? **Ans: a**

- a) 1200 W
- b) 1000 W
- c) 800 W
- d) 750 W
- e) No power saving

14. A heat engine operating in a cycle receives **6 kW** of heat from a reservoir at **500 K** and rejects heat to a sink at **300 K**, delivering **2 kW** of work. Which of the following statements is correct? **Ans: d**

- a) The heat engine is not possible.
- b) It is a reversible heat engine.
- c) The irreversibility associated with it is 4 kW
- d) The second law efficiency is 0.83

15. An ideal gas goes through a constant temperature reversible heat addition process. Identify the incorrect statement. **Ans: d**

- a) Internal energy and enthalpy remain constant
- b) Entropy increases
- c) Volume increases
- d) Pressure increases

16. Which of the following is not a valid statement? **Ans: c**

- a) $\oint p dv = \oint T ds$ is valid for reversible cycles undergone by the closed system or steady flow system.

- b) In a nonideal adiabatic compressor, the exit entropy is always greater than the inlet entropy.
- c) For an irreversible compression process the exit entropy must be greater than the inlet value.
- d) Like heat transfer, work transfer is also responsible for entropy change in an actual process.
17. Three copper blocks in a rigid insulated container are initially at different temperatures. As they exchange heat to come to the same temperature, their stored energy change by **-260 kJ**, **+240 kJ** and **+100 kJ**. Nothing else is in the container except air. Determine the ΔE of the air. **Ans: c**
- a) -30 kJ
- b) -50 kJ
- c) -80 kJ
- d) -6.67 kJ
- e) No change
18. Which of the following energies may or may not be a part of internal energy? **Ans: c**
- a) Molecular potential energy
- b) Chemical energy
- c) Solid distortion energy
- d) Gravitational potential energy
19. During the free expansion of an ideal gas in an insulated chamber, which of the following statements is correct? **Ans: c**
- a) Internal energy increases
- b) $\int p dv = 0$
- c) $\int T ds > 0$
- d) The entropy of the gas decreases as the pressure decreases.
20. Which of the following statements is correct? **Ans: e**
- a) During the phase change of a pure substance from saturated liquid to saturated vapour at constant pressure, the work done $= w = p_{sat}(v_g - v_f)$
- b) During the phase change of a pure substance from saturated liquid to saturated vapour at constant pressure, the heat supplied $= q = T_{sat}(s_g - s_f)$
- c) At the critical point, the latent heat of vaporization is zero.
- d) Evaporation occurs when the vapour pressure of a liquid is less than its saturation pressure corresponding to its temperature.
- e) All of the above
21. Subcooled liquid and a binary mixture of air and superheated vapour can remain in thermodynamic equilibrium. This statement is **Ans: b**
- (a) Wrong
- (b) Correct
- (c) Partially Correct
- (d) Difficult to comment.

22. Water vapor can never exist at -10°C . This statement is **Ans: b**
- Partially Correct
 - Wrong
 - Correct
 - Difficult to comment.
23. The surface temperature of liquid water in a large lake is 20°C . The possible range of the pressure exerted by the evaporated vapor at **11 AM** on the lake surface is **Ans: b & d**
- More than 2.338 kPa.
 - Less than 2.258 kPa.
 - Less than 2.551 kPa but more than 2.338 kPa.
 - Less than 2.338 kPa but more than 2.15 kPa.

v_f (m^3/kg)	v_g (m^3/kg)	u_f (kJ/kg)	u_g (kJ/kg)	h_f (kJ/kg)	h_g (kJ/kg)
0.001002	57.7897	83.9376	2402.91	83.94	2538.06

Table 1: The thermodynamic properties of the saturated liquid and vapor at 20°C .

24. Energy required to vaporize saturated liquid water while keeping pressure constant at **100 kPa** is. **Ans: c**
- More than 2260 kJ/kg but less than 2270 kJ/kg.
 - Less than 2100 kJ/kg but more than 2050 kJ/kg.
 - Less than 2260 kJ/kg but more than 2200 kJ/kg.
 - Less than 2050 kJ/kg but more than 2000 kJ/kg.

P_{sat} (kPa)	T_{sat} ($^{\circ}\text{C}$)	v_f (m^3/kg)	v_g (m^3/kg)	u_f (kJ/kg)	u_g (kJ/kg)
100	99.62	0.001043	1.69400	417.33	2506.06

Table 2: The thermodynamic properties of the saturated liquid and vapor at 100 kPa.

25. Saturated water vapor at 70°C is to be condensed as it flows through a tube at a rate of **0.35 kg/s**. The condensate leaves the tube as a saturated liquid at 50°C . The rate of heat transfer from the tube is **Ans: c**
- Approximately 817 kJ/s.
 - Approximately 834 kJ/s.
 - Approximately 846 kJ/s.
 - Approximately 907 kJ/s.

$T_{sat} (^{\circ}\text{C})$	$P_{sat} (\text{kPa})$	$v_f (\text{m}^3/\text{kg})$	$v_g (\text{m}^3/\text{kg})$	$u_f (\text{kJ/kg})$	$u_g (\text{kJ/kg})$
70	31.19	0.001023	5.04217	292.93	2469.55
50	12.35	0.001012	12.0318	209.30	2443.47

Table 3: The thermodynamic properties of the saturated liquid and vapor at 70°C and 50°C.

26. Air at **1 MPa** and **300°C** is accelerated by a nozzle steadily and adiabatically from a very low velocity ($V \approx 0$) at the inlet to a velocity of **250.0 m/s** at a rate of **50 kg/s**. The exit pressure is **0.1 MPa** and the nozzle is having an isentropic efficiency of $\eta_N = 0.85$. Assume air as a calorically perfect ideal gas with Gas Constant **R = 287.05 J/kg.K** and the ratio of specific heats $\gamma = c_p/c_v = 1.4$. The exit area of the nozzle is **Ans: b**
- Approximately 0.41 m²
 - Approximately 0.31 m²
 - Approximately 0.51 m²
 - Approximately 0.61 m²
27. A household refrigerator with a COP = **1.2** removes heat from the refrigerated space at a rate of **60 kJ/min**. The electric power consumed by the refrigerator is **Ans: c**
- 1.53 kW
 - 1.20 kW
 - 0.83 kW
 - 2.20 kW
28. A Carnot heat pump operates to keep a room warm in Siberia such that the temperature in the room remains at a temperature of **25°C**. The heat pump consumes **500 W** of power when operating and has a COP of **8.5**. The temperature outside the room is **Ans: c**
- 11.05°C
 - 12.03°C
 - 10.05°C
 - 9.01°C
29. Air is compressed from room conditions to a specified pressure in a reversible manner by two compressors: one isothermal and the other adiabatic. If the entropy changes of air Δs_{isot} during the reversible isothermal compression, and Δs_{adia} during the reversible adiabatic compression, the correct statement regarding entropy change of air per unit mass is **Ans: d & f**
- $\Delta s_{isot} = \Delta s_{adia} = 0$
 - $\Delta s_{isot} = \Delta s_{adia} > 0$
 - $\Delta s_{adia} > 0$
 - $\Delta s_{isot} < 0$
 - $\Delta s_{adia} < 0$
 - $\Delta s_{adia} = 0$

30. Consider a steady flow internally irreversible device producing or consuming work with a single stream of fluid flow. The specific entropies of the fluid at the inlet and exit of the device are s_1 and s_2 , respectively. If $s_1 = s_2$, then **Ans: c**
- (a) The device is insulated
 - (b) The device is externally heated
 - (c) The device is externally cooled
 - (d) No such conclusions can be made
31. Pure conductive heat transfer is taking place through a plane wall of stainless steel steadily at a rate of **600 W**. The inner and outer surface temperatures of the wall are **70°C** and **20°C**, respectively. Then considering the wall as the control mass, which of the following statements is/are correct: **Ans: d&g**
- (a) As net entropy of the system is not increasing, net entropy transfer to the system is zero.
 - (b) The net entropy transfer to the system is positive.
 - (c) No entropy is getting generated inside the system, as net entropy of the system is not increasing.
 - (d) The net entropy transfer to the system is negative.
 - (e) Entropy is getting generated inside the system, because of external irreversibility.
 - (f) There are no internal irreversibility within the system.
 - (g) Entropy is getting generated inside the system, because of internal irreversibility.
32. Heat is getting generated because of the passage of steady electric current through a conductor, which is losing heat to the ambient steadily due to convective heat transfer mechanisms. Then considering the conductor as the control mass, which of the following statement(s) is/are correct: **Ans: b**
- (a) As net entropy of the system is not increasing, net entropy transfer to the system is zero.
 - (b) The net entropy transfer to the system is negative.
 - (c) No entropy is getting generated inside the system, as net entropy of the system is not increasing.
 - (d) Entropy is getting generated inside the system, because of external irreversibility.
 - (e) The net entropy transfer to the system is positive, which is due to non-mechanical electric work transfer.
 - (f) There is no internal irreversibility within the system.
 - (g) Entropy generation within the system is partially due to non-mechanical electric work transfer and partially due to convective heat transfer to the ambient.
33. When stream flows through a turbine steadily producing work, the fluid essentially **Ans: a**

- (a) Expands
 - (b) Compresses
 - (c) Undergoes an increase in the specific volume
 - (d) Undergoes an increase in temperature
34. A compressor is compressing air (working fluid), which is essentially working as a steady flow device. The work input requirement for the compressor can be minimized by **Ans: b**
- (a) Keeping the compressor walls insulated
 - (b) Cooling the compressor walls
 - (c) Heating the compressor walls
 - (d) None of the above
35. Air is flowing through a single-stage turbine inside a jet-engine. The work output from the turbine can be maximized by **Ans: c**
- (a) Keeping the turbine walls insulated
 - (b) Cooling the turbine walls
 - (c) Heating the turbine walls
 - (d) None of the above
36. A compressor is compressing air from 100 kPa to 1.0 MPa, while working as a steady flow device. The inlet temperature of air is 20°C and the mass-flow rate is $\dot{m} = 1.5$ kg/s. Assume air as a calorically perfect ideal gas with Gas Constant $R = 287.05$ J/kg.K and the ratio of specific heats $\gamma = c_p / c_v = 1.4$. The theoretical minimum work input required for the compressor is **Ans: a**
- (a) 290.50 kw
 - (b) 410.95 kw
 - (c) 354.10 kw
 - (d) 190.50 kw
37. A unit mass of an ideal gas undergoes a reversible isothermal process from pressure P_1 to pressure P_2 while losing heat to the surroundings at temperature T in the amount of q . If the gas constant of the gas is R , the entropy change of the gas s during this process is **Ans: f**
- (a) $\Delta s = R \ln(P_2 / P_1)$
 - (b) $\Delta s = R \ln(P_2 / P_1) - q/T$
 - (c) $\Delta s = R \ln(P_1 / P_2)$
 - (d) $\Delta s = R \ln(P_1 / P_2) - q/T$
 - (e) $\Delta s = 0$
 - (f) $\Delta s = -q/T$
38. Consider a simple compressible fluid is flowing through a single-stage turbine, which is producing work output while working as a steady flow irreversible device. The fluid expands to pressure P_2 at the exit from the inlet pressure P_1 . Let, $\Delta h_s = (h_1 - h_{2s})$ and $\Delta h_a = (h_1 - h_2)$. Here, h_1 and h_2 are specific enthalpies at the inlet and exit respectively, while,

h_{2s} is the specific enthalpy if isentropic expansion takes place from inlet pressure P_1 to exit pressure P_2 . It is postulated that $\Delta h_s > \Delta h_a$. This postulate may be wrong **Ans: c**

- (a) When the turbine is completely insulated
- (b) When the turbine is externally heated
- (c) When the turbine is externally cooled
- (d) The postulate is always correct because of internal irreversibility

39. Consider air is flowing steadily through a two-stage single-stage compressor with inter-cooling. Assume air behaves as a calorically perfect ideal gas and the compression process may be idealized as following a reversible polytropic process. For least work consumption, theoretically the intermediate pressure should be **Ans: b**

- (a) Arithmetic mean of inlet and exit pressures
- (b) Geometric mean of inlet and exit pressures
- (c) Harmonic mean of inlet and exit pressures
- (d) No specific correlation exists

40. The slope of the isobaric lines, for a simple compressible substance in the $(h - s)$ plot (h is the specific enthalpy and s is the specific entropy), i.e., $\left(\frac{\partial h}{\partial s}\right)_P$ is

Ans: c

- (a) Negative and the value is equal to $(-T)$
- (b) Negative and the value is equal to $(-P)$
- (c) Positive and the value is equal to T
- (d) Positive and the value is equal to $1/T$
- (e) Positive and the value is equal to P
- (f) Zero slope
- (g) Infinite slope